



Ergodic observables in non-ergodic systems: The example of the harmonic chain

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ABSTRACT

In the framework of statistical mechanics the properties of macroscopic systems are deduced starting from the laws of their microscopic dynamics. One of the key assumptions in this procedure is the ergodic property, namely the equivalence between time averages and ensemble averages. This property can be proved only for a limited number of systems; however, as proved by Khinchin (1949), weak forms of it hold even in systems that are not ergodic at the microscopic scale, provided that extensive observables are considered.

Here we show in a pedagogical way the validity of the ergodic hypothesis, at a *practical* level, in the paradigmatic case of a chain of harmonic oscillators. By using analytical results and numerical computations, we provide evidence that this non-chaotic integrable system shows ergodic behavior in the limit of many degrees of freedom. In particular, the Maxwell–Boltzmann distribution turns out to fairly describe the statistics of the single particle velocity. A study of the typical time-scales for relaxation is also provided.

1. Introduction

Since the seminal works by Maxwell, Boltzmann and Gibbs, statistical mechanics has been conceived as a link between the microscopic world of atoms and molecules and the macroscopic one where everyday phenomena are observed [1]. The same physical system can be described, in the former, by an enormous number of degrees of freedom N (of the same order of the Avogadro number) or, in the latter, in terms of just a few thermodynamics quantities. Statistical mechanics is able to describe in a precise way the behavior of these macroscopic observables, by exploiting the knowledge of the laws for the microscopic dynamics and classical results from probability theory. Paradigmatic examples of this success are, for instance, the possibility to describe the probability distribution of the single-particle velocity in an ideal gas [1,2], as well as the detailed behavior of phase transitions [3,4] and critical phenomena [5,6]. In some cases (Bose–Einstein condensation [7], absolute negative temperature systems [8–10]) the results of statistical mechanics were able to predict states of the matter that were never been observed before.

In spite of the above achievements, a complete consensus about the actual reasons for such a success has not been yet reached within the statistical mechanics community. The main source of disagreement is the so-called “ergodic hypothesis”, stating that time averages (the ones actually measured in physics experiments) can be computed as ensemble averages (the ones appearing in statistical mechanics calculations). Specifically, a system is called ergodic when the value of the time average of any observable is the same as the one obtained by taking the average over the energy surface, using the microcanonical distribution [11]. In this respect,

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it is worth mentioning that the word “Ergode” was originally introduced by Boltzmann in order to refer to the microcanonical ensemble [12]. For a discussion on the original meaning of the “ergodic hypothesis” and its evolution over time, see Refs. [13,14].

From a mathematical point of view, ergodicity holds only for a small amount of physical systems: the KAM theorem [15–17] establishes that non-trivial dynamics cannot be ergodic in general. Nonetheless, the ergodic hypothesis happens to work extremely well also for many of these systems, meaning that it provides results in perfect agreement with the numerical and experimental observations [18–20]. Non-ergodicity is seldom observed in many-particle systems, and it is typically the signature of a glassy behavior, or of out-of-equilibrium dynamics [21,22]. Different explanations for this behavior have been provided. Without going into the details of the controversy, three main points of view can be identified: (i) the “classical” school based on the seminal works by Boltzmann and the important contribution of Khinchin, where the main role is played by the presence of many degrees of freedom in the considered systems [23–28]; (ii) those, like the *Prigogine* school, who recognize in the chaotic nature of the microscopic evolution the dominant ingredient [29–32]; (iii) the *maximum entropy* point of view, which does not consider statistical mechanics as a physical theory but as an inference methodology based on incomplete information [33–36].

The main aim of the present contribution is to clarify, at a pedagogical level, how ergodicity manifests itself for some relevant degrees of freedom, in non-ergodic systems. We say that ergodicity occurs “at a practical level”. To this end, a classical chain of N coupled harmonic oscillators turns out to be an excellent case of study: being an integrable system, it cannot be suspected of being chaotic; still, “practical” ergodicity is recovered for relevant observables, in the limit of $N \gg 1$. We believe that this kind of analysis supports the traditional point of view of Boltzmann, which identifies the large number of degrees of freedom as the reason for the occurrence of ergodic behavior for physically relevant observables. Of course, these conclusions are not new. In the works of Khinchin (and then Mazur and van der Linden) [23,37–40] it is rigorously shown that the ergodic hypothesis holds for observables that are computed as an average over a finite fraction of the degrees of freedom, in the limit of $N \gg 1$. Specifically, if we limit our interest to this particular (but non-trivial) class of observables, the ergodic hypothesis holds for almost all initial conditions (but for a set whose probability goes to zero for $N \rightarrow \infty$), within arbitrary accuracy. In addition, several numerical results for weakly non-linear systems [41–43], as well as integrable systems [16,44], present strong indications of the poor role of chaotic behavior, implying the dominant relevance of the many degrees of freedom. In this sense the concept of ergodicity is strictly related to that of “typicality” (of the trajectories), extensively studied in the literature [45].

Despite these ideas are well established, we think it may be useful, especially from a pedagogical point of view, to analyze the explicit example of the harmonic chain, where analytical calculations can be made (to some extent) without losing physical intuition about the model. In particular, we are able to explicitly compute the empirical distribution of the single-particle momentum and the distribution of momenta at a given time, given an out-of-equilibrium initial condition where only a small fraction of the normal modes is excited. Even if many results are available for the thermalization of the considered model [46–48] and its quantum counterpart [49], this computation is, to the best of our knowledge, a novel contribution.

The rest of this paper is organized as follows. In Section 2 we briefly recall basic facts about the chosen model, to fix the notation and introduce some formulae that will be useful in the following. Section 3 contains the main result of the paper. We present an explicit calculation of the empirical distribution of the single-particle momentum, given a system starting from out-of-equilibrium initial conditions. We show that in this case the Maxwell–Boltzmann distribution is an excellent approximation in the $N \rightarrow \infty$ limit. Section 4 is devoted to an analysis of the typical times at which the described ergodic behavior is expected to be observed; a comparison with a noisy version of the model (which is ergodic by definition) is also provided. In Section 5 we draw our final considerations.

2. Model

We are interested in the dynamics of a one-dimensional chain of N classical harmonic oscillators of mass m . The state of the system is described by the canonical coordinates $\{q_j(t), p_j(t)\}$ with $j = 1, \dots, N$; here $p_j(t)$ identifies the momentum of the j th oscillator at time t , while $q_j(t)$ represents its position. The j th and the $(j+1)$ -th particles of the chain interact through a linear force of intensity $\kappa|q_{j+1} - q_j|$, where κ is the elastic constant. We will assume that the first and the last oscillator of the chain are coupled to virtual particles at rest, with infinite inertia (the walls). The case of a “ring” geometry, where both walls are at the same point on the real line ($q_0 \equiv q_{N+1} \equiv 0$), is considered.

The Hamiltonian of the model reads therefore

$$\mathcal{H}(\mathbf{q}, \mathbf{p}) = \sum_{j=0}^N \frac{p_j^2}{2m} + \sum_{j=0}^N \frac{m\omega_0^2}{2} (q_{j+1} - q_j)^2, \quad (1)$$

where $\omega_0 = \sqrt{\frac{\kappa}{m}}$.

Such a system is integrable and, therefore, trivially non-ergodic. This can be easily seen by considering the normal modes of the chain [47], i.e. the set of canonical coordinates

$$Q_k = \sqrt{\frac{2}{N+1}} \sum_{j=1}^N q_j \sin\left(\frac{jk\pi}{N+1}\right) \quad (2a)$$

$$P_k = \sqrt{\frac{2}{N+1}} \sum_{j=1}^N p_j \sin\left(\frac{jk\pi}{N+1}\right), \quad (2b)$$

with $k = 1, \dots, N$. Let us notice that the $\{Q_k\}$ and the $\{P_k\}$ have the physical dimensions of positions and momenta, respectively. Indeed, by rewriting the Hamiltonian in terms of these new canonical coordinates one gets [47]

$$\mathcal{H}(\mathbf{Q}, \mathbf{P}) = \frac{1}{2} \sum_{k=1}^N \left(\frac{P_k^2}{m} + \omega_k^2 Q_k^2 \right), \quad (3)$$

where the frequencies of the normal modes are given by

$$\omega_k = 2\omega_0 \sin \left(\frac{\pi k}{2N+2} \right). \quad (4)$$

In other words, the system can be mapped into a collection of independent harmonic oscillators with characteristic frequencies $\{\omega_k\}$. This system is clearly non-ergodic, as it admits N integrals of motion, namely the energies

$$E_k = \frac{1}{2} \left(\frac{P_k^2}{m} + \omega_k^2 Q_k^2 \right)$$

associated to the normal modes.

In spite of its apparent simplicity, the above system allows the investigation of some nontrivial aspects of the ergodic hypothesis, and helps clarifying the physical meaning of this assumption.

3. Ergodic behavior of the momenta

In this section we analyze the statistics of the single-particle momenta of the chain. We aim to show that they approximately follow a Maxwell–Boltzmann distribution

$$\mathcal{P}_{MB}(p) = \sqrt{\frac{\beta}{2\pi m}} e^{-\beta p^2/2m} \quad (5)$$

in the limit of large N , where β is the inverse temperature of the system. With the chosen initial conditions, $\beta = N/E_{tot}$. Firstly, extending some classical results by Kac [46,48], we focus on the empirical distribution of the momentum of one particle, computed from a unique long trajectory, namely

$$\mathcal{P}_e^{(j)}(p) = \frac{1}{T\alpha} \int_0^T dt \delta \left(\frac{p - p_j(t)}{\alpha} \right). \quad (6)$$

Here α is a constant with the physical dimensions of a momentum. Then we consider the marginal probability distribution $\mathcal{P}_e(p, t)$ computed from the momenta $\{p_j\}$ of all the particles at a specific time t , i.e.

$$\mathcal{P}_e(p, t) = \frac{1}{N\alpha} \sum_{j=1}^N \delta \left(\frac{p - p_j(t)}{\alpha} \right). \quad (7)$$

In both cases we assume that the system is prepared in an atypical initial condition. More precisely, we consider the case in which $Q_j(0) = 0$, for all j , and the total energy E_{tot} , at time $t = 0$, is equally distributed among the momenta of the first N^* normal modes, with $1 \ll N^* \ll N$:

$$P_j(0) = \begin{cases} \sqrt{2mE_{tot}/N^*} & \text{for } 1 \leq j \leq N^* \\ 0 & \text{for } N^* < j \leq N. \end{cases} \quad (8)$$

In this case, the dynamics of the first N^* normal modes is given by

$$\begin{aligned} Q(t) &= \sqrt{\frac{2E_{tot}}{\omega_k^2 N^*}} \sin(\omega_k t) \\ P(t) &= \sqrt{\frac{2mE_{tot}}{N^*}} \cos(\omega_k t). \end{aligned} \quad (9)$$

Let us stress that the considered initial condition is quite pathological, meaning that it is far from the typical equilibrium configurations of the chain. Our aim is to show that even starting from such an atypical condition, single-particle observables thermalize to their expected equilibrium state.

3.1. Empirical distribution of single-particle momentum

Our aim is to compute the empirical distribution of the momentum of a given particle p_j , i.e., the distribution of its values measured in time. This analytical calculation was carried out rigorously by Mazur and Montroll in Ref. [47]. Here, we provide an alternative argument that has the advantage of being more concise and intuitive, in contrast to the mathematical rigor of [47]. Our approach exploits the computation of the moments of the distribution; by showing that they are the same, in the limit of infinite measurement time, as those of a Gaussian, it is possible to conclude that the considered momentum follows the equilibrium Maxwell–Boltzmann distribution. The assumption $N \gg 1$ will enter explicitly the calculation.

The momentum of the j th particle can be written as a linear combination of the momenta of the normal modes by inverting Eq. (2b):

$$\begin{aligned} p_j(t) &= \sqrt{\frac{2}{N+1}} \sum_{k=1}^N \sin\left(\frac{jk\pi}{N+1}\right) P_k(t) \\ &= 2\sqrt{\frac{mE_{tot}}{(N+1)N^*}} \sum_{k=1}^{N^*} \sin\left(\frac{kj\pi}{N+1}\right) \cos(\omega_k t) \end{aligned} \quad (10)$$

where the ω_k 's are defined by Eq. (4), and the dynamics (9) has been taken into account. The n th empirical moment of the distribution is defined as the average $\overline{p_j^n}$ of the n th power of p_j over a measurement time T :

$$\begin{aligned} \overline{p_j^n} &= \frac{1}{T} \int_0^T dt p_j^n(t) \\ &= \frac{1}{T} \int_0^T dt (C_{N^*})^n \prod_{l=1}^n \left[\sum_{k_l=1}^{N^*} \sin\left(\frac{k_l j \pi}{N+1}\right) \cos(\omega_{k_l} t) \right] \\ &= (C_{N^*})^n \sum_{k_1=1}^{N^*} \dots \sum_{k_n=1}^{N^*} \sin\left(\frac{k_1 j \pi}{N+1}\right) \dots \sin\left(\frac{k_n j \pi}{N+1}\right) \frac{1}{T} \int_0^T dt \cos(\omega_{k_1} t) \dots \cos(\omega_{k_n} t) \end{aligned} \quad (11)$$

with

$$C_{N^*} = 2\sqrt{\frac{mE_{tot}}{(N+1)N^*}}. \quad (12)$$

We want to study the integral appearing in the last term of the above equation. To this end, let us assume that $\omega_1, \omega_2, \dots, \omega_q$ are incommensurable, meaning that their ratios cannot be expressed as rational numbers, which is typically the case. Provided that T is much larger than $(\omega_j - \omega_k)^{-1}$ for each choice of $1 \leq k < j \leq q$, a well known result [46] assures that

$$\frac{1}{T} \int_0^T dt \cos^{r_1}(\omega_1 t) \dots \cos^{r_q}(\omega_q t) \simeq \frac{1}{(2\pi)^q} \int_0^{2\pi} d\theta_1 \dots \int_0^{2\pi} d\theta_q \cos^{r_1}(\theta_1) \dots \cos^{r_q}(\theta_q). \quad (13)$$

The relation can be intuitively understood by considering a particle moving in a q -dimensional hypercube with constant velocity $(\omega_1, \dots, \omega_q)$. Let the hypercube have side 2π and periodic boundary conditions: because of the incommensurability of the $\{\omega_j\}$, the trajectory is composed by parallel straight lines crossing the hypercube in a uniform fashion. Taking the temporal average of a function with periodicity 2π along every direction is therefore equivalent to a uniform average over the whole hypercube.

By recalling now that

$$\frac{1}{2\pi} \int_0^{2\pi} d\theta \cos^r(\theta) = \begin{cases} \frac{(r-1)!!}{r!!} & \text{for } r \text{ even} \\ 0 & \text{for } r \text{ odd,} \end{cases} \quad (14)$$

one immediately gets

$$\frac{1}{T} \int_0^T dt \cos^{r_1}(\omega_1 t) \dots \cos^{r_q}(\omega_q t) \simeq \frac{(r_1-1)!!}{r_1!!} \dots \frac{(r_q-1)!!}{r_q!!} \quad \text{if } r_1, \dots, r_q \text{ are all even.} \quad (15)$$

Instead, if at least one of the $\{r_j\}$ is odd, the above quantity vanishes, again with corrections due to the finite time T . Since the smallest frequency is ω_1 , one has that the error is at most of the order $O(q(\omega_1 T)^{-1}) \simeq O(qN/\omega_0 T)$.

Let us consider again the integral in the last term of Eq. (11). The ω_k 's are, in general, incommensurable. Therefore, the integral vanishes when n is odd, since in that case at least one of the $\{n_l\}$, $l = 1, \dots, q$, will be odd. When n is even, the considered quantity is different from zero as soon as the k 's are pairwise equal, so that $n_1 = \dots = n_q = 2$. In the following we will neglect the contribution of terms containing groups of four or more equal k 's: if $n \ll N^*$, the number of these terms is indeed $\sim O(N^*)$ times less numerous than the pairings, and it can be neglected if $N^* \gg 1$ (which is one of our assumptions on the initial condition). Calling Ω_n the set of possible pairings for the vector $\mathbf{k} = (k_1, \dots, k_l)$, we have then

$$\overline{p_j^n} \simeq \left(\frac{C_{N^*}}{\sqrt{2}} \right)^n \sum_{\mathbf{k} \in \Omega_n} \prod_{l=1}^n \sin\left(\frac{k_l j \pi}{N+1}\right), \quad (16)$$

with an error of $O(1/N^*)$ due to neglecting groups of 4, 6 and so on, and an error $O(nN/\omega_0 T)$ due to the finite averaging time T , as discussed before. Factor $2^{-n/2}$ comes from the explicit evaluation of Eq. (13).

At fixed j , we need now to estimate the sums appearing in the above equation, recalling that the k 's are pairwise equal. If $j \gg \frac{N}{N^*}$, the arguments $\varphi_l = \frac{k_l j \pi}{N+1}$ of the periodic functions can be thought of as if independently extracted from a uniform distribution $\mathcal{P}(\varphi_l) = \frac{1}{2\pi}$ in the interval $[0, 2\pi)$, as far as this computation is concerned. Indeed, one has that $\varphi_l \gg \frac{k_l \pi}{N^*}$, which is a quantity of order $O(1)$ for most of the considered $\{k_l\}$; φ_l is therefore much larger than 2π , and its phase with respect to this periodicity can be assumed to be random (but for the negligible set of values of k_l that are simple integer factors of $N+1$).

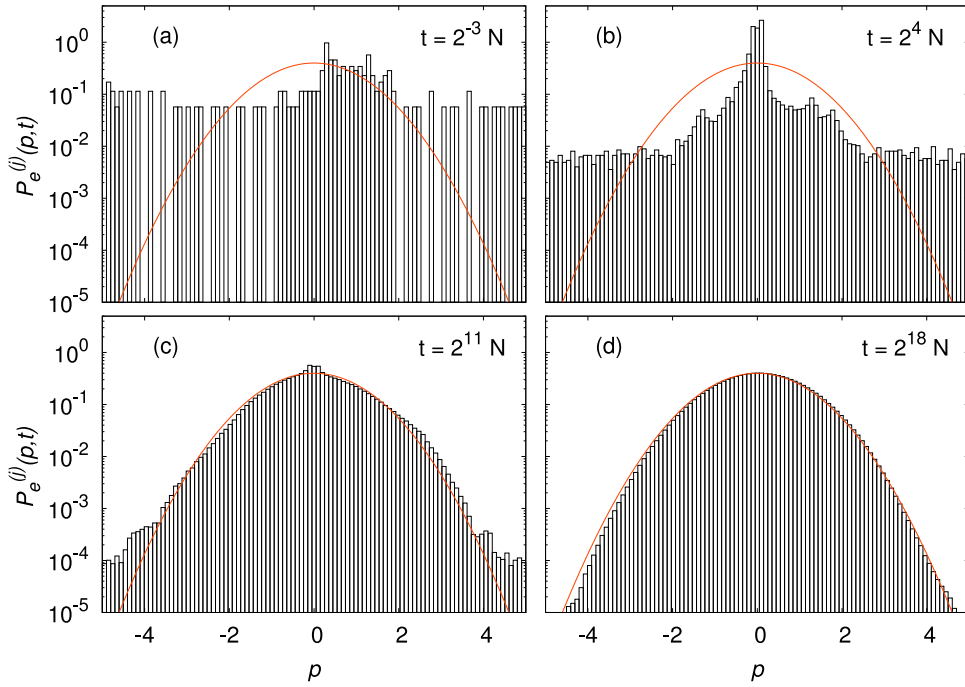


Fig. 1. Convergence of the empirical distribution of the single particle momentum. The four panels show histograms for the momentum of the j th particle, computed by considering the value of p_j at integral times $0, 1, 2, \dots, t$. The system is prepared at time 0 in the atypical initial condition described in the text, where only N^* normal modes are excited. Notice that only one trajectory is considered (there is no average over an ensemble of initial conditions). Different values of t are considered: when the total time is large enough, the distribution approaches the Maxwell-Boltzmann equilibrium (red curve). Here $N = 10^3$, $N^* = 0.1N$, $E_{tot} = N$, $\omega_0 = 1$, $j = 123$.

One has:

$$\left\langle \sin^2 \left(\frac{kj\pi}{N+1} \right) \right\rangle \simeq \sum_{k=1}^{N^*} \frac{1}{N^*} \sin^2 \left(\frac{kj\pi}{N+1} \right) \simeq \frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta \sin^2(\theta) = \frac{1}{2}, \quad (17)$$

and

$$\left\langle \prod_{l=1}^n \sin \left(\frac{k_l j \pi}{N+1} \right) \right\rangle \simeq 2^{-n/2}, \quad (18)$$

if $\mathbf{k} \in \Omega_n$. As a consequence

$$\overline{p_j^n} \simeq \left(\frac{C_{N^*}}{2} \right)^n (N^*)^{n/2} \mathcal{N}(\Omega_n) \simeq \left(\frac{mE_{tot}}{N+1} \right)^{n/2} \mathcal{N}(\Omega_n), \quad (19)$$

where

$$\mathcal{N}(\Omega_n) = \begin{cases} (n-1)!! & \text{for } n \text{ even} \\ 0 & \text{for } n \text{ odd} \end{cases}$$

is the number of ways in which we can choose the pairings. These are the moments of a Gaussian distribution with zero average and $\frac{mE_{tot}}{N+1}$ variance.

Summarizing, it is possible to show that, if $n \ll N^* \ll N$, the first n moments of the distribution are those of a Maxwell-Boltzmann distribution. In the limit of $N \gg 1$ with N^*/N fixed, the Gaussian distribution is thus recovered up to an arbitrary number of moments. Let us note that the assumption $Q_j(0) = 0$, while allowing to make the calculations clearer, is not really relevant. Indeed, if $Q_j(0) \neq 0$ we can repeat the above computation while replacing $\omega_k t$ by $\omega_k t + \phi_k$, where the phases ϕ_k take into account the initial conditions.

Fig. 1 shows the standardized histogram of the relative frequencies of single-particle velocities of the considered system, in the $N \gg 1$ limit, with the initial conditions discussed before. As expected, the shape of the distribution tends to a Gaussian in the large-time limit.

3.2. Distribution of momenta at a given time

A similar strategy can be used to show that, at any given time t large enough, the histogram of the momenta is well approximated by a Gaussian distribution. Again, the large number of degrees of freedom plays an important role. We want to compute the empirical moments

$$\langle p^n \rangle(t) = \frac{1}{N} \sum_{j=1}^N p_j^n(t), \quad (20)$$

defined according to the distribution $\mathcal{P}_e^{(j)}(p)$ introduced by Eq. (6). Using again Eq. (10) we get

$$\begin{aligned} \langle p^n \rangle(t) &= \frac{1}{N} \sum_{j=1}^N (C_{N^*})^n \left[\sum_{k=1}^{N^*} \sin\left(\frac{kj\pi}{N+1}\right) \cos(\omega_k t) \right]^n \\ &= \frac{1}{N} (C_{N^*})^n \sum_{k_1=1}^{N^*} \dots \sum_{k_n=1}^{N^*} \left[\prod_{l=1}^n \cos(\omega_{k_l} t) \right] \sum_{j=1}^N \sin\left(\frac{k_1 j \pi}{N+1}\right) \dots \sin\left(\frac{k_n j \pi}{N+1}\right). \end{aligned} \quad (21)$$

Reasoning as before, we see that the sum over j vanishes in the large N limit unless the k 's are pairwise equal. Again, we neglect the terms including groups of 4 or more equal k 's, assuming that $n \ll N^*$, so that their relative contribution is $O(1/N^*)$. That sum selects paired values of k for the product inside the square brackets, and we end with

$$\langle p^n \rangle(t) \simeq \frac{1}{N} (C_{N^*})^n \sum_{\mathbf{k} \in \Omega_n} \left[\prod_{l=1}^n \cos(\omega_{k_l} t) \right]. \quad (22)$$

If t is “large enough” (we will come back to this point in the following section), different values of ω_{k_l} lead to completely uncorrelated values of $\cos(\omega_{k_l} t)$. Hence, as before, we can consider the arguments of the cosines as extracted from a uniform distribution, obtaining

$$\langle p^n \rangle(t) \simeq \left(\frac{C_{N^*}}{2} \right)^n (N^*)^{n/2} \mathcal{N}(\Omega_n) \simeq \left(\frac{mE_{tot}}{N+1} \right)^{n/2} \mathcal{N}(\Omega_n). \quad (23)$$

These are again the moments of the equilibrium Maxwell–Boltzmann distribution. We had to assume $n \ll N^*$, meaning that a Gaussian distribution is recovered only in the limit of large number of degrees of freedom.

The empirical distribution can be compared with the Maxwell–Boltzmann by looking at the Kullback–Leibler divergence [50] $K[\mathcal{P}_e(p, t), \mathcal{P}_{MB}(p)]$ which provides a sort of distance between the empirical $\mathcal{P}_e(p, t)$ and the Maxwell–Boltzmann:

$$K[\mathcal{P}_e(p, t), \mathcal{P}_{MB}(p)] = - \int \mathcal{P}_e(p, t) \ln \frac{\mathcal{P}_{MB}(p)}{\mathcal{P}_e(p, t)} dp. \quad (24)$$

Fig. 2 shows how the Kullback–Leibler divergences approach their equilibrium limit, for different values of N . As expected, the transition happens on a time scale that depends linearly on N .

A comment is in order: even if this behavior may look similar to the H-Theorem for diluted gases, such a resemblance is only superficial. Indeed, while in the cases of diluted gases the approach to the Maxwell–Boltzmann is due to the collisions among different particles that actually exchange energy and momentum, in the considered case the “thermalization” is due to a dephasing mechanism.

4. Analysis of the time scales

In the previous section, when considering the distribution of the momenta at a given time, we had to assume that t was “large enough” in order for our approximations to hold. In particular we required $\cos(\omega_{k_1} t)$ and $\cos(\omega_{k_2} t)$ to be uncorrelated as soon as $k_1 \neq k_2$. Such a dephasing hypothesis amounts to asking that

$$|\omega_{k_1} t - \omega_{k_2} t| > 2\pi c, \quad (25)$$

where c is the number of phases by which the two oscillator have to differ before they can be considered uncorrelated. The constant c may be much larger than 1, but it is not expected to depend strongly on the size N of the system. In other words, we require

$$t > \frac{c}{|\omega_{k_1} - \omega_{k_2}|} \quad (26)$$

for each choice of k_1 and k_2 . To estimate this typical relaxation time, we need to pick the minimum value of $|\omega_{k_1} - \omega_{k_2}|$ among the possible pairs (k_1, k_2) . This term is minimized when $k_1 = \bar{k}$ and $k_2 = \bar{k} - 1$ (or vice-versa), with $\bar{k}$ chosen such that $\omega_{\bar{k}} - \omega_{\bar{k}-1}$ is minimum. In the large- N limit this quantity is approximated by

$$\omega_{\bar{k}} - \omega_{\bar{k}-1} = \omega_0 \sin\left(\frac{\bar{k}\pi}{2N+2}\right) - \omega_0 \sin\left(\frac{\bar{k}\pi - \pi}{2N+2}\right) \simeq \omega_0 \cos\left(\frac{\bar{k}\pi}{2N+2}\right) \frac{\pi}{2N+2}, \quad (27)$$

which is minimum when $\bar{k}$ is maximum, i.e. for $\bar{k} = N^*$.

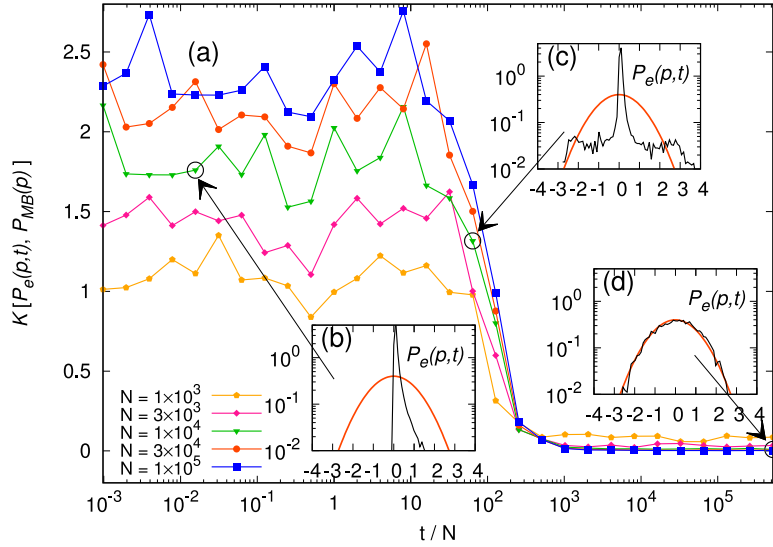


Fig. 2. Kullback–Leibler divergence between the empirical distribution of the momenta (7) and the Maxwell–Boltzmann (5), as a function of time. Panel (a) shows the behavior of $K[P_e(p,t), P_{MB}(p)]$, defined by Eq. (24), for different values of N . As before, the system is initialized at time $t = 0$ in a far-from-equilibrium state where only a small fraction of the modes is excited. The transition to equilibrium takes place on a time scale that is proportional to N , as expected from the argument in the main text. Panels (b)–(d) show the detail of the empirical distribution $P_e(p,t)$ at different times, for the $N = 10^4$ case. Unlike in Fig. 1, here the histogram is built including the momenta of *all* particles in the system, at the considered time. Parameters: $N^* = 0.1N$, $E_{tot} = N$, $\omega_0 = 1$.

Dephasing is thus expected to occur at

$$t > \frac{4cN}{\omega_0 \cos\left(\frac{N^*\pi}{2N}\right)}, \quad (28)$$

i.e. $t > 4cN/\omega_0$ in the $N^*/N \ll 1$ limit.

It is instructive to compare this characteristic time with the typical relaxation time of the “damped” version of the considered system. In order to do so, we assume that our chain of oscillators is now in contact with a viscous medium which acts at the same time as a thermal bath and as a source of viscous friction. By considering the (stochastic) effect of the medium, one gets the Klein–Kramers stochastic process [51,52]

$$\begin{aligned} \frac{\partial q_j}{\partial t} &= \frac{p_j}{m} \\ \frac{\partial p_j}{\partial t} &= \omega_0^2(q_{j+1} - 2q_j + q_{j-1}) - \gamma p_j + \sqrt{2\gamma T} \xi_j \end{aligned} \quad (29)$$

where γ is the damping coefficient and T is the temperature of the thermal bath (we are taking the Boltzmann constant k_B equal to 1). Here the $\{\xi_j\}$ are time-dependent, delta-correlated Gaussian noises such that $\langle \xi_j(t) \xi_k(t') \rangle = \delta_{jk} \delta(t - t')$. Such a system is surely ergodic and the stationary probability distribution is the familiar equilibrium one

$$P_s(\mathbf{q}, \mathbf{p}) \propto e^{-\frac{H(\mathbf{q}, \mathbf{p})}{T}}. \quad (30)$$

Also in this case we can consider the evolution of the normal modes. By taking into account Eqs. (2) and (29) one gets

$$\begin{aligned} \dot{Q}_k &= \frac{1}{m} P_k \\ \dot{P}_k &= -\omega_k^2 Q_k - \frac{\gamma}{m} P_k + \sqrt{2\gamma T} \zeta_k \end{aligned} \quad (31)$$

where the $\{\zeta_k\}$ are again delta-correlated Gaussian noises. It is important to notice that also in this case the motion of the modes is independent (i.e. the friction does not couple normal modes with different k); nonetheless, the system is ergodic, because the presence of the noise allows it to explore, in principle, any point of the phase-space.

The Fokker–Planck equation for the evolution of the probability density function $\mathcal{P}(Q_k, P_k, t)$ of the k th normal mode can be derived using standard methods [51]:

$$\partial_t \mathcal{P} = -\partial_{Q_k} (P_k \mathcal{P}) + \partial_{P_k} \left(\omega_k^2 Q_k \mathcal{P} + \frac{\gamma}{m} P_k \mathcal{P} \right) + \gamma T \partial_{P_k}^2 \mathcal{P}. \quad (32)$$

The above equation allows to compute also the time dependence of the correlation functions of the system in the stationary state. In particular one gets

$$\frac{d}{dt} \langle Q_k(t) Q_k(0) \rangle = \frac{1}{m} \langle P_k(t) Q_k(0) \rangle \quad (33)$$

and

$$\frac{d}{dt} \langle P_k(t) Q_k(0) \rangle = -\omega_k^2 m \langle Q_k(t) Q_k(0) \rangle - \frac{\gamma}{m} \langle P_k(t) Q_k(0) \rangle, \quad (34)$$

which, once combined together, lead to

$$\frac{d^2}{dt^2} \langle Q_k(t) Q_k(0) \rangle + \frac{\gamma}{m} \frac{d}{dt} \langle Q_k(t) Q_k(0) \rangle + \omega_k^2 \langle Q_k(t) Q_k(0) \rangle = 0. \quad (35)$$

For $\omega_k < \gamma/m$ the solution of this equation admits two characteristic frequencies $\tilde{\omega}_\pm$, namely

$$\tilde{\omega}_\pm = \frac{\gamma}{2m} \left(1 \pm \sqrt{1 - \frac{m^2 \omega_k^2}{\gamma^2}} \right). \quad (36)$$

In the limit $\omega_k \ll \gamma/m$ one has therefore

$$\tilde{\omega}_- \simeq \frac{m}{4\gamma} \omega_k^2 \simeq \frac{m \omega_0^2 \pi^2 k^2}{\gamma N^2}. \quad (37)$$

Therefore, as a matter of fact, even in the damped case the system needs a time that scales as N^2 in order to get complete relaxation for the modes. As we discussed before, the dephasing mechanism that guarantees for “practical” ergodicity in the deterministic version is instead expected to occur on time scales of order $O(N)$.

5. Conclusions

The main aim of this paper was to expose, at a pedagogical level, some aspects of the foundation of statistical mechanics, namely the role of ergodicity for the validity of the statistical approach to the study of complex systems.

We analyzed a chain of classical harmonic oscillators (i.e. a paradigmatic example of integrable system, which cannot be suspected to show chaotic behavior). This classical model is quite important in the context of condensed matter, as it reproduces the essential features of a 1-dimensional crystal lattice [53,54]. By extending some well-known results by Kac [46], we showed that the Maxwell–Boltzmann distribution approximates with arbitrary precision (in the limit of large number of degrees of freedom) the empirical distribution of the momenta of the system, after a dephasing time which scales with the size of the chain. This is true also for quite pathological initial conditions, where only a small fraction of the normal modes is excited at time $t = 0$. The scaling of the typical dephasing time with the number of oscillators N may appear as a limit of our argument, since this time will diverge in the thermodynamic limit; on the other hand one should consider, as explicitly shown before, that the damped version of this model (which is ergodic by definition) needs times of the order $O(N^2)$ to reach thermalization for each normal mode.

This comparison clearly shows that the effective thermalization observed in large systems has little to do with the mathematical concept of ergodicity, and it is instead related to the large number of components concurring to define the global observables that are usually taken into account (in our case, the large number of normal modes that define the momentum of a single particle). When these components cease to be in phase, the predictions of statistical mechanics start to be effective; this can be observed even in integrable systems, without need for the mathematical notion of ergodicity to hold.

In other words, we believe that the present work gives further evidence of the point of view (which had been substantiated mathematically by the results of Khinchin, Mazur and van der Linden) that the most relevant ingredient of statistical mechanics is the large number of degrees of freedom, and the global nature of the observables that are typically taken into account.

CRedit authorship contribution statement

Marco Baldovin: Conceptualization, Formal analysis, Visualization, Writing – original draft. **Raffaele Marino:** Software, Visualization, Writing – original draft, Writing – review & editing. **Angelo Vulpiani:** Conceptualization, Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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